

From Feature Suppression to Feature Reliance: Diagnosing Shape, Texture, and Color Dependence under Natural Distribution Shift

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Abstract

Deep visual classifiers often fail under natural distribution shift, yet the feature-level causes of such failures remain difficult to measure. A common explanation is that ImageNet-trained models rely excessively on texture, but recent controlled-suppression studies suggest that this picture is incomplete: models may depend on color, texture, or shape in ways that vary across architectures and domains. We present a diagnostic framework for measuring *feature reliance* by applying controlled image transformations that selectively suppress color, texture, or shape while tracking accuracy, representation similarity, and predictive distribution shift. We evaluate ResNet-50 and ViT-B/16 on ImageNet, an ImageNet-200 subset aligned with ImageNet-R, and ImageNet-R. Across 2,376 evaluation rows from 66 perturbation settings, shape-disrupting transformations cause the largest reliance scores, especially under ImageNet-R style shift, while texture suppression is often less damaging than expected. ViT-B/16 is more stable than ResNet-50 under moderate perturbations, but both models suffer severe failures under local geometric warping. We further outline a sparse-autoencoder validation protocol for testing whether hidden ViT features expose disentangled reliance factors. These results support feature reliance as a practical lens for diagnosing out-of-distribution robustness and for designing targeted data-centric interventions.

1 Introduction

Modern image classifiers can achieve high in-distribution accuracy while remaining brittle under domain shift. The brittleness is not only a property of the final decision boundary: it often reflects which visual evidence the model uses. If a classifier relies on fragile cues such as local texture, it may fail when the same object appears in a different style or rendering. Conversely, if it relies on shape cues, it may be more robust to some texture shifts but sensitive to geometric disruption.

The long-standing “texture bias” hypothesis argues that ImageNet-trained CNNs prefer local texture over global shape [4]. However, recent work revisiting feature reliance through controlled suppression reports a more nuanced story: measured reliance

depends strongly on how visual attributes are suppressed [2]. This motivates a practical question for robustness research: can we diagnose which feature families a model relies on before designing out-of-distribution (OOD) interventions?

We study this question using controlled transformations that suppress or perturb color, texture, and shape. Given an original image x and transformed image $\tilde{x}_f$ suppressing feature family f , we measure performance degradation relative to chance-corrected original accuracy, together with representation and output changes. Our experiments compare ResNet-50 and ViT-B/16 on ImageNet, ImageNet-200, and ImageNet-R. The ImageNet-200 split uses the ImageNet-R class subset, allowing us to compare in-distribution and natural style-shifted evaluation over the same label space.

The empirical pattern is clear. Color removal is consistently mild. Texture suppression through bilateral filtering is also moderate, and on ImageNet-R it can be nearly neutral for ResNet-50. By contrast, shape-disrupting patch shuffle, patch rotation, and local warping induce the largest failures, particularly on ImageNet-R. The result is not that texture is irrelevant, but that robustness failures cannot be explained by a one-dimensional texture-bias story. Feature reliance must be measured along multiple axes.

Our contributions are:

- We implement a controlled feature-reliance evaluation pipeline for color, texture, and shape perturbations across ImageNet, ImageNet-200, and ImageNet-R.
- We report architecture- and domain-specific reliance patterns for ResNet-50 and ViT-B/16 using accuracy, relative accuracy, Jensen-Shannon divergence, and centered kernel alignment (CKA).
- We introduce a sparse-autoencoder (SAE) validation protocol for connecting input-level suppression to latent feature factors in ViT activations.

2 Related Work

Texture and shape bias. Geirhos et al. showed that ImageNet-trained CNNs often classify images using texture cues and that increasing shape bias can improve robustness [4]. Follow-up work questioned whether texture bias is an intrinsic property of ImageNet-trained CNNs or an artifact of how feature reliance is probed [2]. Our work follows the controlled suppression perspective and extends it to natural style-shift evaluation.

Out-of-distribution generalization. OOD robustness benchmarks such as ImageNet-R evaluate whether classifiers generalize from natural photographs to renditions, cartoons, paintings, and other style-shifted images [6]. Domain generalization work has also emphasized that simple data-centric interventions can be competitive with more complex algorithmic changes [5]. We treat feature reliance as a diagnostic step before choosing such interventions.

Representation similarity and mechanistic validation. CKA provides a stable measure of representation similarity across neural networks and transformations [8]. Sparse autoencoders have recently become a common tool for discovering interpretable latent directions in large models [7]. In vision, concept-level interpretability has been studied through unit-concept alignment, open-world concept discovery, and sparse-autoencoder analysis of visual representations [1, 9, 10]. Our SAE validation protocol asks whether reliance factors measured at the image level are reflected in sparse latent activations inside a ViT, and whether such factors can ultimately support targeted representation-level control [3].

3 Method

3.1 Controlled Feature Suppression

For each image x , we construct transformed images $\tilde{x}_f$ using deterministic or stochastic transformations targeting a visual feature family f :

- **Color:** grayscale conversion with full color removal.
- **Texture:** bilateral filtering, which smooths local texture while preserving many edges.
- **Shape:** patch shuffle, patch rotation, and local geometric warping.

The implementation also records image-level suppression diagnostics: local variance ratio (LV), high-frequency energy ratio (HFE), edge SSIM (ESSIM), and gradient correlation (GC). Texture score is the mean of LV and HFE; shape score is the mean of ESSIM and GC. Lower texture scores indicate stronger texture suppression, while lower shape scores indicate stronger shape disruption.

3.2 Feature Reliance Score

Let A_0 be the model accuracy on original images, A_f the accuracy after suppressing feature family f , and $A_c = 1/C$ the chance accuracy for C classes. We compute chance-corrected relative accuracy:

$$R_f = \frac{A_f - A_c}{A_0 - A_c}. \quad (1)$$

Feature reliance is then:

$$\text{FR}_f = 1 - R_f. \quad (2)$$

A value near zero means the model remains accurate after suppressing f ; a value near one means the model collapses toward chance. We also compute Jensen-Shannon divergence between original and transformed logits and linear CKA between original and transformed representations.

3.3 Sparse Autoencoder Validation

To connect image-level reliance to latent features, we train a ReLU sparse autoencoder on ViT-B/16 block activations. The current implementation collects patch tokens from a target ViT block, normalizes token activations with streaming statistics, initializes the decoder bias using a geometric median, and trains an overcomplete dictionary with an L1 activation penalty:

$$\mathcal{L}_{SAE} = \|x - \hat{x}\|_2^2 + \lambda \|z\|_1, \quad (3)$$

where $z = \text{ReLU}((x - b_{dec})W_{enc} + b_{enc})$ and $\hat{x} = zW_{dec} + b_{dec}$. The validation script measures reconstruction MSE, normalized MSE, cosine similarity, active latent counts, latent frequency, and label entropy. In the next experimental iteration, we will compare SAE latent activation shifts under each suppression family to test whether sparse factors separate color-, texture-, and shape-sensitive concepts.

4 Experimental Setup

Models. We evaluate two pretrained models from `timm`: ResNet-50 with ImageNet-1K weights and ViT-B/16 with augmented ImageNet-1K weights. Each model uses its corresponding preprocessing statistics and resize/crop protocol.

Datasets. We evaluate on ImageNet validation, ImageNet-200, and ImageNet-R. ImageNet-200 is constructed by selecting the 200 ImageNet classes that overlap with ImageNet-R. This allows direct comparison between in-distribution images and natural OOD renditions with the same class space.

Perturbation sweep. We run 66 perturbation settings. Bilateral filtering sweeps sigma-color and sigma-space values; local warping sweeps warp amplitude and smoothing; patch shuffle and patch rotation use a fixed grid setting in the current result table. Each setting evaluates original, grayscale, bilateral, patch shuffle, patch rotation, and local warp variants. The resulting analysis table contains 2,376 evaluation rows.

5 Results

5.1 Baseline Accuracy

Table 1 reports original-image accuracy. Both models perform similarly on ImageNet and ImageNet-200, while ImageNet-R causes a large natural OOD drop.

5.2 Feature Reliance Differs by Feature Family

Table 2 summarizes mean feature reliance across perturbation settings. Color removal produces small reliance scores, with grayscale FR ranging from 0.0577 to 0.1479. Bilateral texture suppression is also modest in most settings and is nearly neutral for ResNet-50 on ImageNet-R. Shape perturbations are substantially more damaging. On

Model	Dataset	Accuracy
ResNet-50	ImageNet	0.8035
ResNet-50	ImageNet-200	0.9393
ResNet-50	ImageNet-R	0.4021
ViT-B/16	ImageNet	0.7914
ViT-B/16	ImageNet-200	0.9333
ViT-B/16	ImageNet-R	0.3812

Table 1: Original-image accuracy before controlled suppression.

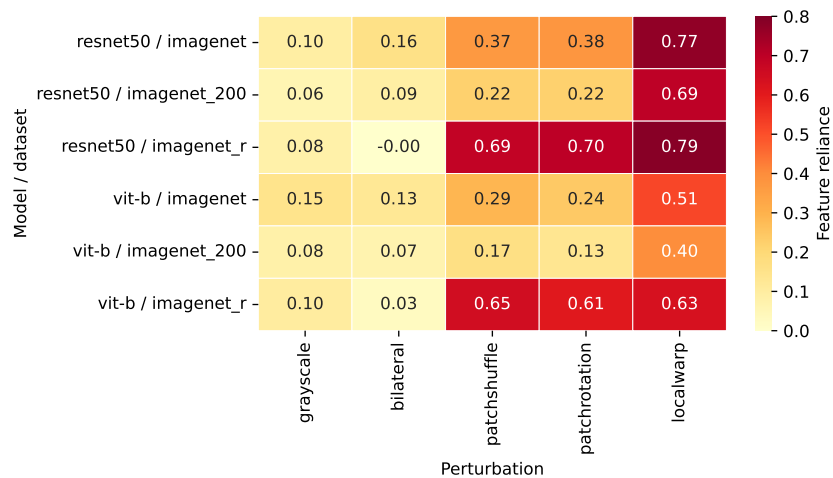


Figure 1: Mean feature reliance across model, dataset, and perturbation family. Shape-disrupting perturbations, especially local warp and patch-level rearrangements, dominate the ImageNet-R failure pattern.

ImageNet-R, patch shuffle and patch rotation produce FR above 0.60 for both models, and local warp reaches 0.7870 for ResNet-50 and 0.6314 for ViT-B/16.

5.3 Representation Shift Tracks Reliance

Feature reliance correlates strongly with output and representation changes. Across non-original evaluations, FR has correlation 0.9349 with Jensen-Shannon divergence and -0.8188 with CKA. Thus, perturbations that cause accuracy collapse also induce larger predictive distribution shifts and lower representation similarity. This supports the interpretation that feature suppression affects internal representations, not merely final-layer calibration.

Model	Dataset	Perturbation	Acc.	Rel. Acc.	FR	CKA	JS
ResNet-50	ImageNet	Grayscale	0.7271	0.9049	0.0951	0.7254	0.0947
ResNet-50	ImageNet	Bilateral	0.6754	0.8404	0.1596	0.6756	0.1356
ResNet-50	ImageNet	Patch shuffle	0.5087	0.6326	0.3674	0.4612	0.2709
ResNet-50	ImageNet	Patch rotation	0.5005	0.6225	0.3775	0.4602	0.2790
ResNet-50	ImageNet	Local warp	0.1856	0.2300	0.7700	0.3060	0.5261
ViT-B/16	ImageNet	Grayscale	0.6745	0.8521	0.1479	0.8363	0.1277
ViT-B/16	ImageNet	Bilateral	0.6923	0.8746	0.1254	0.8768	0.1034
ViT-B/16	ImageNet	Patch shuffle	0.5623	0.7102	0.2898	0.7933	0.2089
ViT-B/16	ImageNet	Patch rotation	0.6004	0.7584	0.2416	0.8163	0.1770
ViT-B/16	ImageNet	Local warp	0.3849	0.4857	0.5143	0.5793	0.3530
ResNet-50	ImageNet-R	Grayscale	0.3699	0.9190	0.0810	0.7722	0.1945
ResNet-50	ImageNet-R	Bilateral	0.4029	1.0021	-0.0021	0.7315	0.2213
ResNet-50	ImageNet-R	Patch shuffle	0.1296	0.3137	0.6863	0.5189	0.4499
ResNet-50	ImageNet-R	Patch rotation	0.1251	0.3024	0.6976	0.5415	0.4216
ResNet-50	ImageNet-R	Local warp	0.0896	0.2130	0.7870	0.4433	0.4995
ViT-B/16	ImageNet-R	Grayscale	0.3418	0.8953	0.1047	0.7368	0.2081
ViT-B/16	ImageNet-R	Bilateral	0.3709	0.9728	0.0272	0.7944	0.1639
ViT-B/16	ImageNet-R	Patch shuffle	0.1376	0.3524	0.6476	0.5959	0.3399
ViT-B/16	ImageNet-R	Patch rotation	0.1523	0.3916	0.6084	0.6495	0.2946
ViT-B/16	ImageNet-R	Local warp	0.1436	0.3686	0.6314	0.5214	0.3554

Table 2: Mean controlled-suppression results. FR denotes feature reliance (1 – relative accuracy); JS denotes Jensen-Shannon divergence.

5.4 ViT-B/16 Is More Stable under Moderate Shape Disruption

ViT-B/16 generally exhibits lower shape reliance than ResNet-50 on ImageNet and ImageNet-200. On ImageNet, patch rotation gives FR 0.2416 for ViT-B/16 versus 0.3775 for ResNet-50, and local warp gives FR 0.5143 versus 0.7700. However, the advantage narrows under ImageNet-R: both architectures become highly sensitive to patch-level shape disruption. This suggests that architectural robustness is conditional on the domain and perturbation family.

5.5 Severe Local Warping Exposes a Failure Mode

The strongest local-warp settings collapse both models near chance. For example, the best reliance setting on ImageNet uses alpha 80 and sigma 0.5, yielding FR 0.9982 for ResNet-50 and 0.9978 for ViT-B/16. These results should not be read as natural robustness estimates, because extreme local warp can produce visually unnatural images. Instead, they expose a useful stress test: models retain high-frequency texture statistics but fail when local geometry is destroyed.

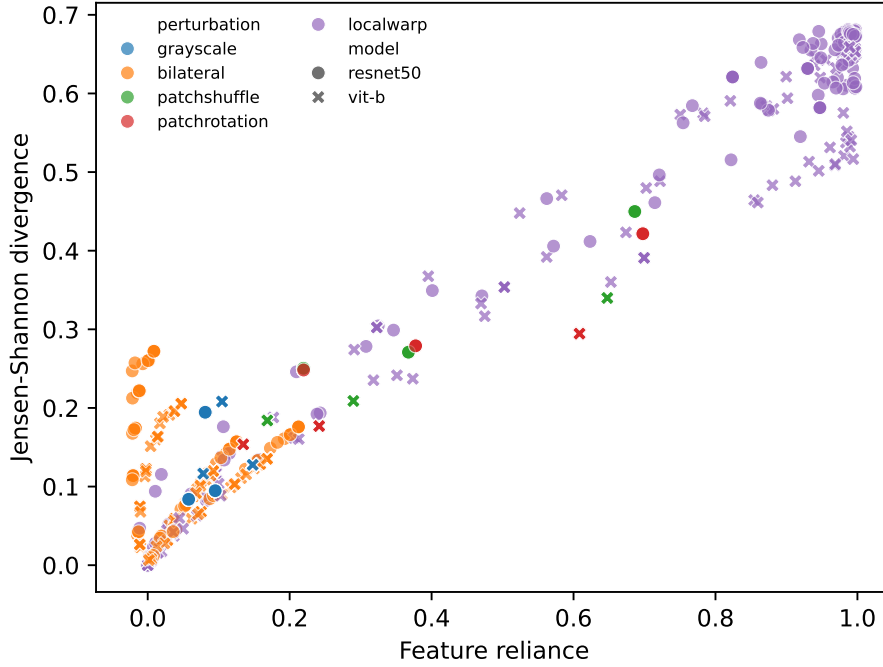


Figure 2: Feature reliance is strongly associated with predictive distribution shift, measured by Jensen-Shannon divergence between original and transformed logits.

6 Discussion

The experiments challenge a simple texture-bias explanation for OOD failure. Texture suppression is not uniformly catastrophic, and on ImageNet-R it can be almost neutral. Shape disruption, by contrast, is consistently severe under style shift. A practical implication is that robustness interventions should be chosen based on measured reliance. If a model is already robust to texture suppression but fragile to local geometry, adding more texture augmentation may not address the dominant failure mode.

The SAE validation pipeline provides a route from diagnosis to intervention. If sparse latent units show selective sensitivity to shape-preserving or shape-disrupting transformations, then feature reliance can be studied not only as an input-level phenomenon but also as a mechanistic property of hidden representations. This opens the possibility of targeted latent editing, regularization, or data augmentation guided by the discovered reliance factors.

7 Limitations

First, controlled suppression is only as valid as the transformation family. Bilateral filtering, patch shuffling, and local warping are proxies for visual feature removal, not

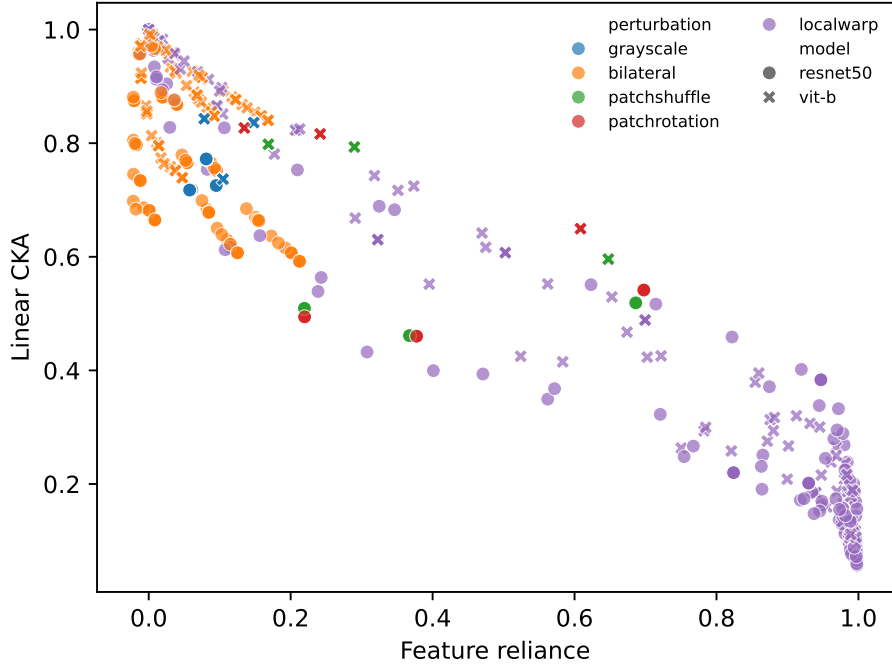


Figure 3: Feature reliance is negatively associated with representation similarity, measured by linear CKA between original and transformed representations.

perfect interventions. Second, the current experiments use two pretrained architectures; broader claims require additional CNNs, transformers, self-supervised models, and CLIP-like models. Third, local warp contains extreme settings that are useful for stress testing but may not represent natural distribution shift. Finally, the SAE component is currently a validation protocol rather than a completed causal intervention; future work should report latent-level perturbation and editing results.

8 Conclusion

We presented a feature-reliance diagnostic framework for measuring how visual classifiers depend on color, texture, and shape under both in-distribution and natural OOD evaluation. Across ResNet-50 and ViT-B/16, controlled suppression shows that shape disruption is the dominant failure mode in our ImageNet-R setting, while texture suppression is often less damaging than expected. These findings suggest that feature reliance is a useful bridge between robustness evaluation and targeted intervention design. The accompanying SAE protocol points toward a mechanistic next step: identifying sparse latent factors that explain and potentially control reliance under distribution shift.

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